Quick Start Guide: Setting Up a Secure Network in AWS with a Virtual Private Cloud

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Who This Guide Is For

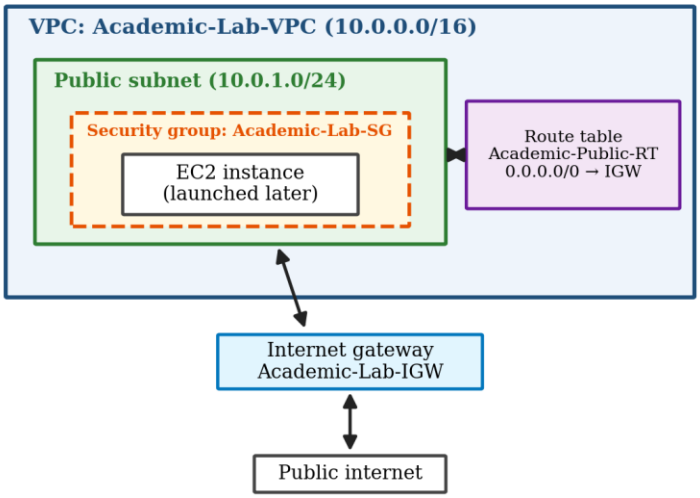
This guide is for Ensign College information technology students who are starting their first cloud computing coursework. Many beginners feel stuck on their first Amazon Web Services (AWS) network because the console offers many options and the parts depend on each other. This guide gives you the exact settings to use, in the right order, so you can build a working network in about 15 to 20 minutes. A well-built network also keeps your lab separate from other projects and limits who can reach your server, which are habits every cloud professional needs.

What You Will Build

When you finish, you will have a Virtual Private Cloud (VPC) with a public subnet, an internet gateway, a route table, and a security group. This is the foundation you need before you launch a server in AWS. Figure 1 shows how the parts connect.

Figure 1

Architecture of the Academic Lab VPC



Before You Begin

Make sure you have the following:

- An AWS account, such as a Free Tier account or an AWS Academy Learner Lab.
- Your sign-in details for the AWS Management Console.
- A Region selected in the console, such as **US East (N. Virginia)**. Use the same Region for every step.
- A basic idea of IPv4 addresses. The block **10.0.0.0/16** is a private range of about 65,000 addresses, and **10.0.1.0/24** is a smaller slice of 256 addresses inside it.

Follow the Five Steps

Figure 2 shows the five steps in order. Complete them in this order, because each step depends on the one before it. For more detail on any setting, see the Amazon VPC User Guide (Amazon Web Services, n.d.).

Figure 2

The Five Steps to Build the Network



Estimated time: about 15–20 minutes

Step 1: Create the VPC

A VPC is your own private network inside AWS. Everything else you build goes inside it.

1. Sign in to the AWS Management Console.
2. Type **VPC** in the search bar and open the VPC service.
3. Choose **Create VPC**.
4. Under **Resources to create**, select **VPC only**.
5. Enter **Academic-Lab-VPC** as the **Name tag**.
6. Select **Manual input** for **IPv4 CIDR block**, then enter **10.0.0.0/16**.
7. Leave **Tenancy** set to **Default**.
8. Choose **Create VPC**.

Step 2: Create a Public Subnet

A subnet is a smaller section of your VPC where servers live. It becomes public only after you finish Step 4.

1. Choose **Subnets** in the left menu, then choose **Create subnet**.
2. For **VPC ID**, select **Academic-Lab-VPC**.
3. Enter **Academic-Public-Subnet-1** as the **Subnet name**.
4. Choose the first **Availability Zone** in the list.
5. Enter **10.0.1.0/24** as the **IPv4 subnet CIDR block**.
6. Choose **Create subnet**.
7. Select the new subnet, then choose **Actions** > **Edit subnet settings**.
8. Select **Enable auto-assign public IPv4 address**, then choose **Save**.

Step 3: Attach an Internet Gateway

An internet gateway is the door between your VPC and the internet. It does nothing until you attach it.

1. Choose **Internet gateways** in the left menu, then choose **Create internet gateway**.
2. Enter **Academic-Lab-IGW** as the **Name tag**, then choose **Create internet gateway**.
3. Choose **Actions** > **Attach to VPC**.
4. Select **Academic-Lab-VPC**, then choose **Attach internet gateway**.

Step 4: Add a Route to the Internet

A route table tells network traffic where to go. This route sends all internet-bound traffic through your gateway.

1. Choose **Route tables** in the left menu, then choose **Create route table**.
2. Enter **Academic-Public-RT** as the **Name**, then select **Academic-Lab-VPC** as the **VPC**.
3. Choose **Create route table**.
4. Select the new route table, open the **Routes** tab, and choose **Edit routes**.
5. Choose **Add route**, then enter **0.0.0.0/0** as the **Destination**.
6. For **Target**, select **Internet Gateway**, then choose **Academic-Lab-IGW**.
7. Choose **Save changes**.
8. Open the **Subnet associations** tab and choose **Edit subnet associations**.
9. Select **Academic-Public-Subnet-1**, then choose **Save associations**.

Step 5: Create a Security Group

A security group works like a firewall for your server by deciding what traffic is allowed in. Table 1 lists the two rules you will add. Never allow SSH from Anywhere. Choose My IP so that only your computer can sign in.

Table 1

Inbound Rules for Academic-Lab-SG

Type	Protocol	Port	Source	Purpose
SSH	TCP	22	My IP	Secure admin access
HTTP	TCP	80	Anywhere-IPv4 (0.0.0.0/0)	Public web access

1. Choose **Security groups** in the left menu, then choose **Create security group**.
2. Enter **Academic-Lab-SG** as the **Security group name**.
3. Enter **Allow SSH from my IP and HTTP from anywhere** as the **Description**.
4. For **VPC**, select **Academic-Lab-VPC**.
5. Under **Inbound rules**, choose **Add rule**. Select **SSH** as the **Type** and **My IP** as the **Source**.

6. Choose **Add rule** again. Select **HTTP** as the **Type** and **Anywhere-IPv4** as the **Source**.
7. Leave the outbound rule at its default setting.
8. Choose **Create security group**.

Check Your Work

Open the VPC console and confirm each item:

- The VPC shows the state **Available** and the CIDR **10.0.0.0/16**.
- **Academic-Public-Subnet-1** belongs to **Academic-Lab-VPC** and has auto-assign public IPv4 turned on.
- **Academic-Lab-IGW** shows the state **Attached**.
- **Academic-Public-RT** has a route from **0.0.0.0/0** to **Academic-Lab-IGW** and is associated with the subnet.
- **Academic-Lab-SG** lists both inbound rules.

Troubleshooting

If something does not work, find your problem in Table 2. Most problems come from a missed setting in an earlier step, so compare your screen with the steps above before you start over.

Table 2

Common Problems and Fixes

Problem	Likely cause	Fix
A new instance has no public IP	Auto-assign is off for the subnet	Choose Actions > Edit subnet settings and turn on auto-assign public IPv4. The setting applies to new instances only.
You cannot connect with SSH	The SSH rule is missing, or your IP address changed	Edit the SSH rule and select My IP again.
There is no internet access	The route is missing or the gateway is not attached	Confirm the route 0.0.0.0/0 to the gateway and the subnet association.
A web page does not load	The HTTP rule is missing	Add the HTTP rule from Table 1.

Clean Up and Next Steps

The VPC, subnet, gateway, route table, and security group are free to keep. Servers and public IP addresses can cost money, so terminate any practice server when you finish. Your next step is to launch an EC2 server in Academic-Public-Subnet-1, select Academic-Lab-SG, and host a simple web page. When you are comfortable with these five steps, try rebuilding the network from memory. Repeating the process once is the fastest way to remember it.

References

Amazon Web Services. (n.d.). *Amazon VPC user guide*. https://docs.aws.amazon.com/vpc/latest/userguide/